UNIT - III

Reinforced Concrete Buildings

Syllabus (as given):

Principles of earthquake resistant design of RC members - structural models for frame buildings - seismic methods of analysis - IS code based methods - vertical irregularities - plan configuration problems - lateral load resisting systems - design lateral forces as per IS 1893 (Part 1):2016 - equivalent lateral force procedure - lateral distribution of base shear.

1. PRINCIPLES OF EQ RESISTANT DESIGN OF RC MEMBE

RC buildings are the most common structures in India. Their performance in an earthquake depends on how well the members are proportioned and detailed to be ductile. Design = (i) get forces from analysis (IS 1893), (ii) design sections (IS 456), (iii) detail for ductility (IS 13920:2016).

- *Strong column - weak beam: sum of column flexural capacities at a joint ≥ 1.4 x sum of beam capacities (IS 13920:2016) so plastic hinges form in beams.*
- *Capacity design for shear: design shear force V_e = shear from gravity + shear from plastic moment capacities at ends ($V_e = V_g \pm 1.4\,(M_u,sag + M_u,hog)/L$) - ensures flexural (ductile) failure occurs before shear (brittle).*
- *Confinement of concrete at member ends (plastic hinge zones) with closely spaced hoops with 135° hooks.*
- *Adequate anchorage and lap splices located away from high-moment zones (columns: central half); hooks inside the confined core.*
- *Joint integrity: beam-column joints must stay elastic; transverse reinforcement continued through joint; beam bars properly anchored.*
- *Under-reinforced sections: tension steel yields before concrete crushes (limit ρ_{max} = 2.5% in beams; columns 0.8%-4%).*
- *Keep structure redundant: many bays of moment frames so that failure of one hinge does not collapse the structure.*

Member	Seismic role	Key IS 13920:2016 provisions
Beam	Dissipates energy by flexural hinging	$b \geq 200$ mm, $b/D \geq 0.3$, $D \leq L/4$; $\rho_{min} = 0.24\sqrt{f_{ck}}/f_y$, ρ_{max} = 2.5%; min 2 bars top & bottom throughout; stirrups with 135° hook, spacing ≤ d/4, 8φ, 100 mm in 2d zone
Column	Must remain elastic (axial + bending)	Min size 300 mm, $b/D \geq 0.4$; 0.8% ≤ ρ ≤ 4%; closely spaced hoops over l_0 at ends; $\Sigma M_c \geq 1.4\,\Sigma M_b$
Joint	Transfers forces	Hoops continued through joint; special confining steel; anchorage beyond joint core
Slab (diaphragm)	Distributes inertia forces	Monolithic with beams; openings < 50% area; adequate chords/collectors

Member	Seismic role	Key IS 13920:2016 provisions
Shear wall	Main lateral load resisting	Min thickness 150 mm; ρ_{min} 0.25% each way; boundary elements if needed

2. STRUCTURAL MODELS FOR FRAME BUILDINGS

- 3-D frame model (space frame) - most accurate; beam and column elements (with rigid end zones), slab as rigid/semi-rigid diaphragm; software like ETABS/STAAD.
- Plane frame model - for regular, symmetric buildings; each frame analysed separately for its share of lateral load.
- Lumped mass shear building model - floors rigid, one lateral DOF per floor (per direction) - basis for modal analysis. Stiffness of each storey $k_i = \Sigma 12EI/h^3$ (columns fixed-fixed).
- Equivalent frame / equivalent strut for infill: masonry infill replaced with a diagonal compression strut (width $w_{ds} = 0.175\,\alpha_h^{-0.4}\,L_{ds}$, IS 1893:2016 cl 7.9.2.2).
- Modelling points: cracked section properties (beams 0.35 I_g, columns 0.7 I_g - IS 1893:2016 cl 6.4.3.1 for deflection/period); rigid joints; fixed vs pinned base; P-Δ; soil flexibility; seismic weight lumped at floor levels (with 25-50% IL).
- Number of modes in dynamic analysis: such that total modal mass participation ≥ 90% in each direction.

> Cracked stiffness
> IS 1893:2016: for RC members effective stiffness (cracked): beams 0.35 EI_g, columns 0.70 EI_g, walls 0.70 EI_g (uncracked), 0.35 (cracked) - for analysis under seismic loading.

3. SEISMIC METHODS OF ANALYSIS (IS CODE BASED)

Method	Applicability (IS 1893:2016 cl. 7.6)
Equivalent static (seismic coefficient)	Regular buildings: Zone II & III up to 90 m, Zone IV & V up to 40 m; irregular buildings up to 12 m (IV, V) or 40 m (II, III)
Dynamic analysis (response spectrum / time history)	All others - tall regular, irregular above limits; mandatory for buildings with vertical irregularity, high torsion, special structures

- Response spectrum method: (1) find mode shapes & periods (eigen analysis), (2) for each mode obtain S_a/g, (3) modal base shear $V_k = A_k W_k$, (4) combine modes with SRSS (well-separated modes) or CQC (closely spaced modes, frequency difference < 10%).
- Time-history: accelerograms scaled to the design spectrum; direct integration (Newmark β) of equations of motion - for important structures.
- Dynamic base shear V_B must be ≥ base shear from empirical period; otherwise scale up all results by V_B/V_B.
- Orthogonal effects: 100% in one direction + 30% in the perpendicular direction (and vice versa), take the worst.

4. VERTICAL IRREGULARITIES

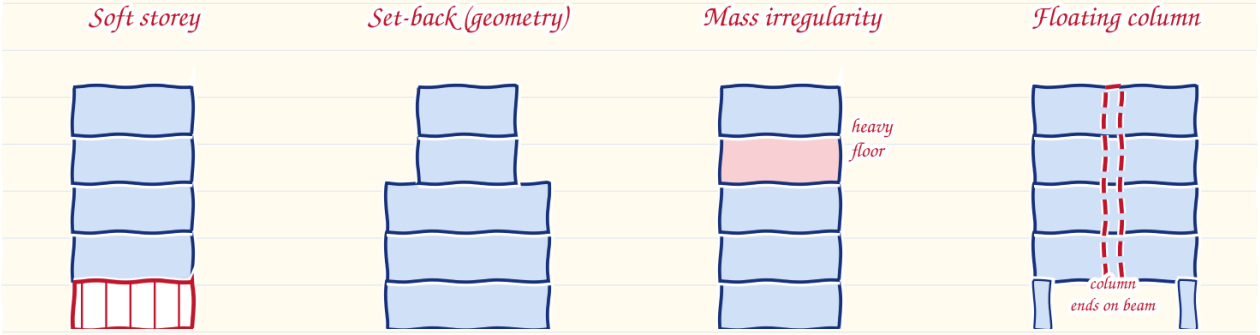


Fig: Common vertical irregularities

- Soft storey (stiffness irregularity) - ground-floor parking, tall ground storey, no infill; drift concentrates -> pancake collapse (Bhuj 2001).
- Weak storey (strength < 80% of above). Mass irregularity (> 150% of neighbouring floors). Vertical geometric irregularity / set-back (> 125%).
- In-plane discontinuity (shear wall not reaching foundation), floating columns (transfer girders, discontinuous load path).
- Short column effect: columns restrained by partial-height infills/mezzanines/split-level floors have a short free length -> very stiff -> attract large shear -> X-shaped shear cracks.
- Consequences: concentration of damage, inelastic demand over limited elements, sudden failure. IS 1893:2016: such buildings need dynamic analysis; soft storey shall be avoided (RC walls added in ground storey).

5. PLAN CONFIGURATION PROBLEMS

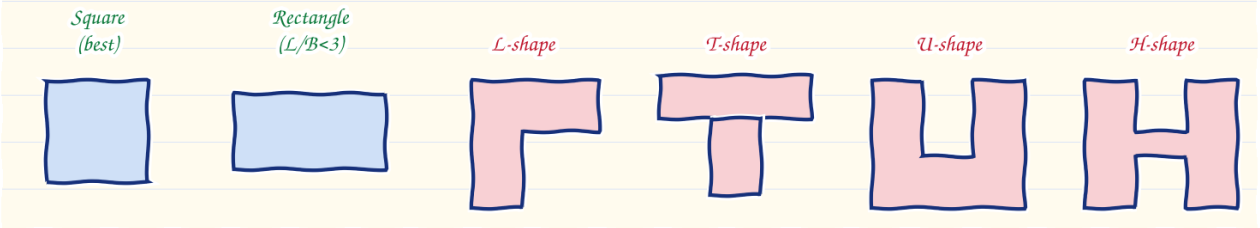


Fig: Plan shapes: good vs poor

- Torsion (CM ≠ CR): corner/edge columns heavily loaded.
- Re-entrant corners (L, T, U, H, + shapes): stress concentration at the inner corner; wings vibrate differently -> use seismic joints.
- Diaphragm discontinuity: large openings (staircase, atrium, lifts) weaken floor action; keep openings < 50% and away from corners/walls, provide trimmer steel.
- Out-of-plane offsets and non-parallel systems: eccentric load paths -> torsion.
- Pounding between adjacent buildings - provide gap; expansion/separation joints in long buildings (> 45-50 m).
- Remedies: symmetric simple shape; distribute walls symmetrically at periphery; limit L/B ≤ 3; separate wings with joints; avoid heavy masses far from CR.

6. LATERAL LOAD RESISTING SYSTEMS

System	How it works	R (IS 1893:2016)	Notes
Ordinary RC moment frame (OMRF)	Frame action, no special detailing	3.0	Zone II only
Special RC moment frame (SMRF)	Ductile frame (IS 13920)	5.0	Zones III-V; up to ~10-12 storeys economical
Ordinary RC shear wall (with OMRF)	Walls resist lateral load	3.0 / 3.5	Stiff; limited ductility
Ductile RC shear wall	Walls detailed per IS 13920	4.0	Efficient 10-40 storeys
Dual (wall + SMRF)	Frames ≥ 25% of base shear designed to share	4.5-5.0	Redundant; best
Steel braced frames, EBF	Axial/shear link yielding	4.0 - 5.0	Steel buildings
Unreinforced masonry (load bearing)	Walls	1.5	Low-rise only (see Unit IV)

- *Moment frames*: flexible, architecturally open; drift governs design; strong-column weak-beam.
- *Shear walls*: resist most lateral load by flexure + shear; reduce drift; need boundary elements; place symmetrically.
- *Dual systems*: walls take the majority in lower storeys, frames take it in the upper storeys; interaction gives nearly uniform drift.
- Floor diaphragms (RC slabs) tie everything together and distribute inertia forces by stiffness.

7. DETERMINATION OF DESIGN LATERAL FORCES - IS 18

Step A. Seismic weight

$W = \Sigma W_i$ with W_i = DL (slab, beams, columns half above/below, walls, finishes) + 25% IL (IL ≤ 3 kN/m²) or 50% IL (IL > 3 kN/m²); roof live load = 0.

Weight of columns/walls in a storey: half above + half below the floor.

Step B. Natural period (empirical, cl. 7.6.2)

RC moment frame (bare, no infill): $T_a = 0.075\ h^{0.75}$

Steel moment frame: $T_a = 0.085\ h^{0.75}$

RC/steel frame with brick infill, shear wall buildings, others: $T_a = 0.09\ h/\sqrt{d}$

h = building height (m) from foundation/ground (excluding basement if walls rigid); d = base dimension (m) in the direction of EQ.

Step C. Design horizontal seismic coefficient

$A_h = (Z/2)(S_a/g) / (R/I)$

S_a/g from the design spectrum for the soil type (I/II/III) at T_a, 5% damping; use $S_a/g = 2.5$ on the plateau.

Code lower bound: $A_h \geq Z/2 \times 1/(R/I) \times 0.25$-type minimum is implicit in $S_a/g = 0.25$ for $T > 4$ s.

Step D. Design base shear

$V_B = A_h W$

8. EQUIVALENT LATERAL FORCE PROCEDURE & LATER

$$Q_i = V_B \cdot W_i h_i^2 / \Sigma W_j h_j^2$$

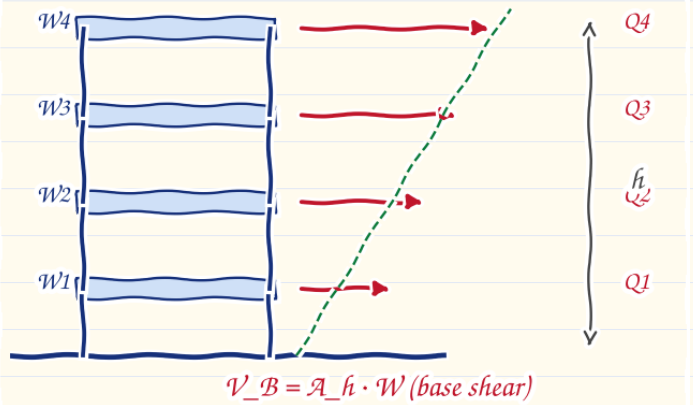


Fig: Lateral distribution of base shear Q_i along height

Vertical distribution (cl. 7.6.3)

$Q_i = V_B \times W_i h_i^2 / \Sigma_{j=1}^{n} W_j h_j^2$

Q_i = design lateral force at floor i; W_i = seismic weight of floor i; h_i = height of floor i measured from base (m).

$\Sigma Q_i = V_B$. Storey shear $V_i = \Sigma$ of Q_j above storey i; overturning moment $M_i = \Sigma Q_j (h_j - h_i)$.

- Lateral force distribution is parabolic (quadratic in height) - assumes fundamental mode shape is linear.
- Horizontal distribution at each floor to frames/walls: (a) rigid diaphragm -> in proportion to relative lateral stiffness (rigidity) of the vertical elements; (b) flexible diaphragm -> by tributary area/mass.
- Torsion: add torsional moment $M_t = Q_i \times e_{di}$ with design eccentricity $e_{di} = 1.5\, e_{si} + 0.05\, b_i$ or $e_{si} - 0.05\, b_i$ (whichever is more severe).
- Storey drift limit: ≤ 0.004 h under design lateral force (with partial factor 1.0), computed with cracked sections; no need of P-Δ if stability index $\theta = P\Delta/(V h) < 0.1$.

- Frame lateral stiffness of one column: $k = 12EI/h^3$ (both ends fixed); sum over columns of storey -> storey stiffness; portal/cantilever methods for approximate analysis of frames.

Worked Example (full procedure)

Problem

G+3 RC school building (Zone IV, $Z = 0.24$; $I = 1.5$ [important]; SMRF $R = 5$; medium soil). Plan 15 m x 12 m, storey height 3.2 m, height $h = 12.8$ m (4 floors), brick infill.

Seismic weights: floor 1,2,3 = 3200 kN each, roof = 2400 kN -> $W = 12\,000$ kN.

Direction of shorter dimension $d = 12$ m: $T_a = 0.09 \times 12.8/\sqrt{12} = 0.332$ s (< 0.55 s) -> $S_d/g = 2.5$

$A_h = (0.24/2) \times 2.5/(5/1.5) = 0.12 \times 2.5/3.333 = 0.09$

$V_B = 0.09 \times 12\,000 = 1080$ kN

Floor	W_i (kN)	h_i (m)	$W_i h_i^2$	$W_i h_i^2/\Sigma$	Q_i (kN)	Storey shear (kN)
Roof (4)	2400	12.8	393 216	0.4615	498.5	498.5
3	3200	9.6	294 912	0.3462	373.9	872.4
2	3200	6.4	131 072	0.1538	166.2	1038.6
1	3200	3.2	32 768	0.0385	41.5	1080.1
Σ	12 000		851 968	1.000	1080	

Check

Overturning moment at base = $\Sigma Q_i h_i = 498.5 \times 12.8 + 373.9 \times 9.6 + 166.2 \times 6.4 + 41.5 \times 3.2 = 6381 + 3589 + 1064 + 133 = 11\,167$ kN.m. Each floor force is then shared among frames in proportion to their stiffness, plus torsion.

9. MORE THEORY YOU NEED FOR THE EXAM

9.1 Response reduction factor R

Actual structures have reserve strength and ductility, so they can be designed for a force much lower than the elastic force. $R = R_\mu \times \Omega \times R_r$ (ductility x over-strength x redundancy).

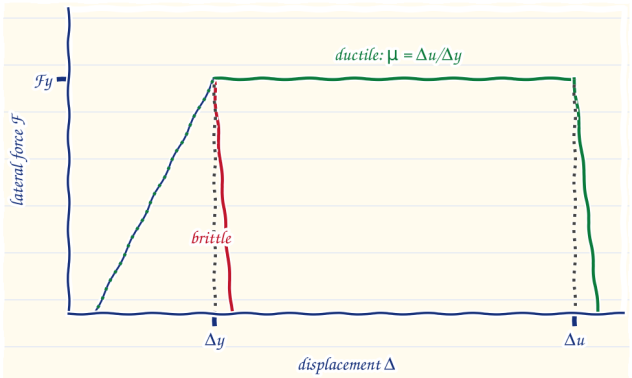


Fig: Elastic force vs design force: reduction by ductility

- Elastic base shear (if structure stayed elastic) $V_{el} = R \times V_{design}$ (e.g., $R = 5$ -> $V_{el} = 5\,V_B$).

- R_μ (ductility reduction): equal-displacement $R_\mu = \mu$ ($T > 0.5$ s); equal-energy $R_\mu = \sqrt{(2\mu - 1)}$ (short period).
- Ω (over-strength) 1.5 - 2.0: actual strength > design strength due to material strength, load factors, minimum steel, slab participation.
- R_r (redundancy): many hinges -> load shared; low redundancy buildings get smaller R.
- Hence IS 1893 gives smaller R to brittle systems (URM 1.5) and larger R to ductile systems (SMRF 5).

9.2 Seismic weight - how to compute (floor example)

Floor 15 m × 12 m (180 m²), live load 3 kN/m², slab 150 mm

Slab 0.15 × 25 = 3.75 kN/m²; floor finish 1.0; partitions 1.0 -> DL = 5.75 kN/m² -> 5.75 × 180 = 1035 kN

Beams (say 0.3 × 0.5 m, total 130 m): 0.3 × 0.35 (below slab) × 25 × 130 = 341 kN

Columns: 20 nos 0.4 × 0.4, storey 3.2 m, half above & below = 3.2 m: 20 × 0.16 × 3.2 × 25 = 256 kN

Outer walls (230 mm, 3.0 m, 54 m perimeter, 19 kN/m³): 0.23 × 3 × 54 × 19 = 708 kN

Live load share: 25% × 3 × 180 = 135 kN (IL ≤ 3 kN/m²)

W_{floor} = 1035 + 341 + 256 + 708 + 135 = 2475 kN (roof: no IL; parapet adds weight).

9.3 Lateral stiffness of columns/frames and distribution to frames

Column stiffness (fixed-fixed, cracked 0.7 EI_g)

$E_c = 5000\sqrt{f_{ck}} = 5000\sqrt{25} = 25\,000$ N/mm² = 25×10^6 kN/m²; column 400 × 400: $I_g = 0.4 \times 0.4^3/12 = 2.133 \times 10^{-3}$ m⁴

$k = 12\,(0.7\,E\,I_g)/h^3 = 12 \times 25e6 \times 1.493e\text{-}3/3.0^3 =$ 16 590 kN/m per column (h = 3 m)

Storey stiffness $K = \Sigma\, k$ of all columns in that storey (parallel springs).

Distribution of floor force to frames (rigid diaphragm, no torsion)

Floor force Q = 360 kN; Frame A, B, C stiffness: 20 000, 30 000, 20 000 kN/m (Σ = 70 000)

$Q_A = 360 \times 20/70 =$ 102.9 kN; $Q_B = 360 \times 30/70 =$ 154.3 kN; $Q_C =$ 102.9 kN -> shares ∝ relative rigidity

Remember

A stiffer element attracts more load. This is why a stiff short column fails first (short-column effect) and why walls take most load in a dual system.

9.4 Torsion - numerical

Eccentricity calculation

Plan 20 m (x) x 10 m; EQ force acts in y-direction, b = 20 m; CM at x = 10.0 m, CR at x = 11.2 m -> e_s = 1.2 m

e_{d1} = 1.5 e_s + 0.05 b = 1.8 + 1.0 = 2.8 m; e_{d2} = e_s - 0.05 b = 1.2 - 1.0 = 0.2 m -> use the worse effect for each element

Torsional moment at the floor M_t = Q_i x e_d = 400 x 2.8 = 1120 kN.m; extra shear in a frame at distance x from CR = M_t k_x x / J, J = Σ $k x^2$ (torsional rigidity).

9.5 Beam-column joint checks (capacity design)

Strong column - weak beam (IS 13920:2016: $\Sigma M_c \geq 1.4 \Sigma M_b$)

At a joint: beam hogging capacity 210 kN.m (left) and sagging 130 kN.m (right): ΣM_b = 340 kN.m

Columns above and below: 280 kN.m each -> ΣM_c = 560 kN.m; required 1.4 x 340 = 476 kN.m -> 560 ≥ 476 OK

Design shear force for beam (capacity design)

Clear span L = 6 m; V_{ug} (gravity, factored 1.2(DL+IL)) = 60 kN; M_u,lim sag = 130 kN.m; M_u,lim hog = 210 kN.m

V_e = V_{ug} + 1.4 (M^{sag} + M^{hog})/L = 60 + 1.4 (130 + 210)/6 = 60 + 79.3 = 139.3 kN (design stirrups for this V_e, not the analysis shear)

9.6 Mode combination in response spectrum method

- *Modal participation factor P_k = Σ $W_i \varphi_{ik}$ / Σ $W_i \varphi_{ik}^2$; modal mass M_k = (Σ $W_i \varphi_{ik}$)²/(g Σ $W_i \varphi_{ik}^2$).*
- *Modal base shear V_k = (Z/2)(S_a/g)$_k$/(R/I) x M_k g; lateral force in mode k at floor i: Q_{ik} = $A_k \varphi_{ik} P_k W_i$.*
- *SRSS: V = √(V_1^2 + V_2^2 + V_3^2 + ...); e.g., V_1 = 800, V_2 = 250, V_3 = 90 kN -> V = √(640 000 + 62 500 + 8 100) = 843 kN*
- *CQC when modal frequencies differ by < 10% (3-D, torsionally coupled buildings): V = √(ΣΣ $\rho_{ij} V_i V_j$).*
- *Include enough modes to capture ≥ 90% of seismic mass in each orthogonal direction.*

9.7 Drift, P-Δ and stability

- *Storey drift Δ_i = u_i - u_{i-1} ≤ 0.004 h_i (e.g., 12.8 mm for h = 3.2 m) under the design lateral force with load factor 1.0 and cracked sections (IS 1893:2016 cl. 7.11.1.1).*
- *P-Δ (second-order) effect: gravity load acting through lateral displacement adds moment; stability index θ = P_u Δ/(V_u h); if θ ≤ 0.1 neglect, θ > 0.1 amplify forces by 1/(1 - θ); θ < 0.33 limit.*
- *Minimum design lateral force: $V_B \geq V_B$; torsion and orthogonal effects included.*

9.8 Dual system - 25% rule and interaction

- In a dual system the moment frames (SMRF) must be designed to resist at least 25% of the design base shear independently, while walls + frames together resist 100%.
- Wall behaves like a vertical cantilever (flexural shape), frame like a shear beam (shear shape): they push and pull each other - walls carry load in lower storeys, frames in upper storeys, giving a more uniform drift profile.
- Wall-frame stiffness ratio influences the distribution; tie them with rigid diaphragms and coupling beams.

9.9 Soft-storey check example

Storey	Lateral stiffness (kN/mm)	% of above	% of avg of 3 above	Result
4	95	-	-	-
3	100	105%	-	OK
2	98	98%	-	OK
1 (ground)	52	53% (< 70%)	52/97.7 = 53% (< 80%)	Soft storey!

Ground storey stiffness 52 kN/mm is < 70% of storey above and < 80% of the average of three above (97.7) -> vertical stiffness irregularity: provide RC shear walls/bracing at ground floor or revise the frame.

9.10 Design process flow (RC building to IS 1893/13920)

1. Fix configuration (regular, symmetric), decide system (SMRF / dual), zone & soil.
2. Preliminary member sizes (beam $b \geq 200$, $D \approx L/12$; column ≥ 300; check against IS 13920 limits).
3. Compute seismic weights; empirical T_a; S_d/g; A_h; V_B; distribute Q_i; torsion.
4. 3-D analysis (ETABS/STAAD) for DL, IL, EL_x, EL_y; check mass participation, drift 0.004 h, torsion irregularity.
5. Combine loads (1.2/1.5/0.9); design beams (flexure + capacity shear), columns (P-M, biaxial), joints, slabs, foundations.
6. Detail as per IS 13920:2016 (hoops, anchorages, laps) - see Unit V.
7. Check non-structural elements, seismic joints, pounding, foundation/liquefaction.

9.11 Lessons from past earthquakes - RC buildings

Earthquake	Typical RC failures observed
Bhuj 2001 (India)	Soft ground storeys (parking), pancake collapse of Ahmedabad multi-storeys, short column failures, poor concrete, weak joints
Kobe 1995	Soft/weak storey at mid-level, inadequate stirrups in columns, shear failure
Northridge 1994	Non-ductile frames, steel connection (weld) fractures, parking structure collapse
Mexico City 1985	Resonance of 6-15 storey buildings with soft-clay site period; pounding
Nepal 2015 (Gorkha)	Unreinforced masonry and mid-rise RC with soft storeys, poor detailing

9.12 Worked example - 2-DOF shear building (response spectrum method)

Data

Two floors, $m_1 = m_2 = 100$ t ($W = 981$ kN each), $k_1 = k_2 = 40$ MN/m; $Z/2 = 0.08$, $R/I = 5$, medium soil.

Eigen-problem: $(K - \omega^2 M)\phi = 0 \Rightarrow \omega^2 = (3 \mp \sqrt{5})/2 \times k/m = 0.382$ k/m and 2.618 k/m, k/m = 400 s^{-2}

$\omega_1 = 0.618 \times 20 = 12.36$ rad/s ($T_1 = 0.508$ s); $\omega_2 = 1.618 \times 20 = 32.36$ rad/s ($T_2 = 0.194$ s)

Mode shapes (floor 1, floor 2): $\phi_1 = (1, 1.618)$; $\phi_2 = (1, -0.618)$

Modal participation, mass and base shear

$P_1 = (1 + 1.618)/(1 + 2.618) = 0.724$; $P_2 = (1 - 0.618)/(1 + 0.382) = 0.276$

Modal mass fraction: $M_1 = (2.618)^2/(2 \times 3.618) = 94.7\%$; $M_2 = (0.382)^2/(2 \times 1.382) = 5.3\%$ ($\Sigma = 100\%$)

Both periods are on the plateau (0.10-0.55 s): $S_d/g = 2.5 \Rightarrow A_h = 0.08 \times 2.5/5 = 0.04$ for both modes

$V_1 = 0.04 \times 0.947 \times 1962 = 74.3$ kN; $V_2 = 0.04 \times 0.053 \times 1962 = 4.1$ kN

SRSS: $V_B = \sqrt{(74.3^2 + 4.1^2)} = 74.4$ kN (compare ELF with same A_h: $0.04 \times 1962 = 78.5$ kN - similar since mode 1 dominates).

Feature	Equivalent lateral force (static)	Response spectrum (dynamic)
Modes considered	Fundamental only	All significant modes (≥ 90% mass)
Force distribution	Parabolic $W h^2$ formula	From mode shapes (can have whiplash at top)
Period	Empirical formula	Computed from model (base shear scaled to empirical-period value)
Torsion, irregular buildings	Approximate	Accurate with 3-D model
Effort	Hand calculation	Software needed

9.13 Common mistakes to avoid

- Forgetting that roof live load is not included and that only 25%/50% of floor IL is included in W.
- Using h from the wrong datum, or d in the wrong direction ($T_a = 0.09\, h/\sqrt{d}$ must be done for each direction).
- Using Z instead of $Z/2$ in A_h.
- Not applying I/R correctly: $A_h = (Z/2)(S_d/g)/(R/I) = (Z/2)(S_d/g) \times I/R$.
- Missing accidental eccentricity 0.05 b and orthogonal 30% rule.
- Using gross sections when code requires cracked stiffness for drift and period.

9.14 Foundations and ground effects on RC buildings

- Isolated footings in Zones III-V (IS 4326 / IS 1893): tie them with plinth/tie beams in both directions so that the foundation moves as one unit; avoid mixing foundation types in one

building.

- Liquefaction: saturated loose sand loses strength (pore pressure rises, effective stress -> 0), buildings tilt/sink, lateral spreading; check with SPT N-values; remedies: densification, stone columns, piles through the liquefiable layer, raft.
- Soft soils amplify motion and lengthen the period (Type III spectrum); foundation stiffness and rocking can alter the period (soil-structure interaction).
- Basement walls: stiff box - reduces h used for T_a (take h from the top of the basement slab only if walls are rigid).
- Pile foundations: design for kinematic effects and lateral loads; pile caps tied by beams.

9.15 Formula sheet - Unit III

Quantity	Formula
Design coefficient	$A_h = (Z/2)(S_a/g)/(R/I)$
Base shear	$V_B = A_h W$
Floor force	$Q_i = V_B W_i h_i^2 / \Sigma W_j h_j^2$
Period (RC frame / infill)	$T_a = 0.075\, h^{0.75}$ / $0.09\, h/\sqrt{d}$
Seismic weight	$W = DL + 0.25\ IL\ (IL \le 3)$ / $0.50\ IL\ (IL > 3)$; roof $IL = 0$
Design eccentricity	$e_d = 1.5\, e_s + 0.05\, b$ or $e_s - 0.05\, b$
Drift limit	$\Delta \le 0.004\, h$
Column stiffness	$k = 12EI/h^3$
Capacity shear (beam)	$V_e = V_{ug} \pm 1.4\,(M_{sag} + M_{hog})/L$
Strong column check	$\Sigma M_c \ge 1.4\, \Sigma M_b$
SRSS	$V = \sqrt{(\Sigma V_k^2)}$

9.16 Approximate analysis of a frame - Portal method (example)

Use when a quick hand check of frame forces under Q_i is needed. Assumptions: point of contraflexure at mid-height of columns and mid-span of beams; interior columns carry twice the shear of exterior columns (for a multi-bay frame).

3-storey, 2-bay frame, storey height 3.2 m; floor loads $Q_3 = 60$, $Q_2 = 40$, $Q_1 = 20$ kN

Storey shears: $V_3 = 60$; $V_2 = 100$; $V_1 = 120$ kN

2 bays -> 3 columns: shear per exterior column = $V/4$, interior = $V/2$: storey 3: 15 / 30 / 15; storey 2: 25 / 50 / 25; storey 1: 30 / 60 / 30 kN

Column end moments = $V_{col} \times h/2 = V_{col} \times 1.6$: storey 3: 24 / 48 / 24; storey 1: 48 / 96 / 48 kN.m

Beam moment at an exterior joint = column moments meeting there (top: $M_{beam} = M_{col}$ top end); at interior joints the sum of column moments is shared equally by the two beams.

Beam shear $V_b = (M_{left} + M_{right})/L_b$; column axial forces from overturning (sum of beam shears).

> *Note*
> *Portal method is accepted only for preliminary design/checks. The code requires a full 3-D analysis, but the idea (storey shear divided among columns by stiffness) is the same.*

QUICK REVISION - Likely questions

1. *State principles of earthquake resistant design of RC members; explain strong column - weak beam and capacity design.*
2. *Describe structural models for frame buildings (shear building, 3-D frame, infill strut).*
3. *Explain IS code based methods of analysis and conditions for dynamic analysis.*
4. *Explain vertical irregularities and plan configuration problems with sketches and remedies.*
5. *Compare lateral load resisting systems: moment frame, shear wall, dual system; R values.*
6. *Calculate design base shear as per IS 1893 (Part 1):2016 and distribute it along the height (numerical).*